

## **Embedded Hardware (HW)**

### **Core Electronics**

- Semiconductor fundamentals
- PN Junction & Diodes
- Rectifiers
- BJT — CB / CE configurations
- MOSFET operation
- Amplifiers
- Oscillators
- Current Mirror circuits

### **Analog & Circuit Analysis**

- KCL / KVL
- Thevenin & Norton Theorems
- Superposition Theorem
- Maximum Power Transfer
- Star-Delta Transformation
- R-2R Ladder
- ADC types & fundamentals
- Transmission Lines

### **Digital Electronics**

- Number systems & Boolean Algebra
- Logic Gates
- K-Maps
- Combinational & Sequential Logic
- Flip-Flops
- Counters & Registers
- FSM
- Moore / Mealy Models

- Sequence Detectors

### **Microcontroller / Processor Basics**

- Microprocessor fundamentals
- Microcontroller fundamentals
- Memory architecture
- Interrupt basics
- Basic peripherals

## **Embedded Software (SW)**

### **C Programming — HIGH PRIORITY**

- C fundamentals
- Data types & operators
- Functions
- Arrays & Strings
- **Pointers & Pointer Arithmetic**
- Structures & Unions
- Enums & Typedef
- Macros & Preprocessor
- static, const, volatile
- Bitwise operations
- Function pointers

### **Memory & Computer Fundamentals**

- Stack vs Heap
- Memory layout
- RAM / ROM / Flash
- Memory allocation
- Microprocessor basics
- Microcontroller basics
- CPU, Registers & Memory Map

### **Embedded C**

- Register manipulation
- Bit manipulation
- GPIO
- Interrupts & ISR
- Timers / Counters
- PWM

- ADC
- Watchdog Timer
- Polling vs Interrupts

### **Communication & RTOS**

- UART
- SPI
- I2C
- Basic RTOS concepts
- Queue
- Scheduling

## **FPGA / RTL**

### **Digital Design**

- Number systems & Boolean Algebra
- Logic Gates
- K-Maps
- Combinational Logic
- Sequential Logic
- MUX / DEMUX
- Encoder / Decoder
- Adders
- Flip-Flops
- Counters
- Registers

### **FSM & State Machines**

- FSM fundamentals
- State diagram & state table
- **Moore Model**
- **Mealy Model**
- Sequence Detectors
- Overlapping / Non-overlapping sequences

### **RTL / Verilog**

- Verilog fundamentals
- Module & Port declarations
- Structural Modeling
- Gate-level Modeling
- Behavioral Modeling
- Dataflow Modeling
- assign

- always
- Blocking vs Non-blocking assignments
- if / else & case
- Testbench basics

## **FPGA & Timing**

- FPGA architecture
- LUTs
- Flip-Flops
- Block RAM
- DSP blocks
- Clock & Reset
- Setup / Hold Time